Example Checklist for Field Sampling Audit

Sampling Organization	Date of Evaluation	Name and Affiliation of Evaluator

Item to be Evaluated	*Y*	*N*	*N/A*	*Comments*
Part 1: Sampling and Analysis Plan (SAP)				
1.1 General Information				
Is there a sampling plan?				
Are there procedures for the transportation, handling, protection, storage, retention and/or disposal of samples, including all provisions necessary to protect the integrity of the sample?				
Is there a documented system for uniquely identifying all samples and subsamples, to ensure that there can be no confusion regarding the identity of such samples at any time?				
Does the sampling process address the factors to be controlled to ensure the validity of the environmental test and calibration results?				
Is there a process for documenting corrective actions taken in the field?				
1.2 Standard Operating Procedures (SOPs)				
Are there SOPs for field activities available at the location where sampling is taking place and are they accessible to all sample collectors?				
Are the SOPs official documents (e.g., signed and dated)?				
Have the SOPs been approved for the project?				
Part 2: Organization, Management & Personnel *(not checked on site)*				
Are the sampling personnel’s qualifications and/or training certifications adequate for the tasks performed?				
Are names of all sampling personnel recorded?				
Do sampling personnel meet minimum qualifications specified in the contract?				
If the sampling organization is supporting a larger organization, are there any arrangements that could cause a conflict of interest?				
Does the sampling agency have policies or procedures to ensure client confidentiality and propriety rights?				
Are staff training records maintained and up to date?				

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Part 3: Equipment				
3.1 General Equipment information				
Is the type of equipment sufficient for the sampling project?				
Is the quantity of equipment sufficient for the sampling project?				
Is the following information recorded for each piece of equipment that will be used for sampling project:				
Maintenance and repair procedures for equipment or instrument?				
Routine cleaning procedures?				
Filling solution replacement for probes?				
Parts replacement for instruments or probes?				
Calendar date for each procedure performed?				
Names of personnel performing maintenance and repair tasks?				
Description of malfunctions associated with any maintenance and repair?				
Vendor service records?				
Inclusive rental dates, types and unique descriptions of rental equipment?				
Is the equipment storage procedure acceptable?				
Is there an existing QC check on sampling equipment?				
3.2 Field Instrument Calibration				
Is information about all calibration standards and reagents used for field testing linked to the calibration information associated with the field testing measurements for the project?				
Are field instruments properly calibrated and calibrations recorded in a bound field log book?				
For each instrument unit used for the sampling project, is the following information recorded for all calibrations:				
Unique identification (designation code) for the instrument?				
Date and time of each calibration and calibration verification?				
Instrument reading or result (display value) for all calibration verifications, with appropriate measurement units?				
Names of analyst performing each calibration of verification?				
Designation of each calibration standard used linked to the associated records for the calibration standard?				
The acceptance criteria for each calibration and verification standard used?				
The assay specifications or acceptance criteria for any QC standard or sample used to independently verify the calibration of the standard?				

Item to be Evaluated	***Y***	***N***	***N/A***	***Comments***
Are all corrective actions performed on the instrument prior to attempting re-verification or recalibration of the instrument linked to the records required for preventive maintenance?				
Does the field instrumentation documentation include the standard concentrations used for calibration?				
Did all field-testing equipment and instrumentation brought to the field appear to function properly?				
Are manufacturer's suggested maintenance activities and any repairs performed and documented for all applicable equipment and instruments?				
3.3 Containers				
Are sample containers well organized, properly prepared, protected from contamination, and ready for use?				
Are proper sample containers and sizes used for each type of sample?				
Are certificates of analysis for pre-cleaned bottles maintained on file?				
Are all containers and container caps free of cracks, chips, discolorations and other features that might affect the integrity of the collected samples?				
3.4 Sampling Equipment				
Is the appropriate equipment used for the sampling project? Check all relevant equipment used for sample collection, handling, storage and transport.				
Is equipment constructed of materials appropriate for the analytes of interest?				
Is equipment brought to the field precleaned?				
For equipment decontaminated on-site in the field, are the date and time of the cleaning procedure recorded in the field records or referenced in an internal SOP?				
Are cleaning steps in all procedures used for decontamination documented either by description or reference to an SOP?				
Are there current maintenance records for all field equipment?				
Part 4: Sampling Event Information				
For all samples, is the following information recorded and transmitted to the client?				
Site name and address?				
Date and time of sample collection?				
Name of sampler responsible for sample transmittal?				
Unique field identification code for each sample container or group of containers?				
Total number of samples collected?				
Required analyses for each sample container or group of containers?				
Sample preservation used for each container or group of containers?				

Item to be Evaluated	*Y*	*N*	*N/A*	*Comments*
Comments about samples, sample sources or other relevant field conditions?				
Identification of common carrier used to transport the samples, when applicable?				
Are shipping invoices and related records from common carriers archived with the field records, when applicable?				
Are sampling locations adequately documented in a bound field log book using indelible ink?				
Are photos taken and is a photo log maintained?				
4.1 Field QC				
Are trip blanks and/or field blanks collected as specified in the approved sampling plan?				
Are field blanks collected after equipment is decontaminated in the field?				
Are field blanks collected if no equipment was cleaned by the sampling organization?				
Are additional samples for matrix spike/matrix spike duplicate analyses collected?				
Are all QC samples collected in the same manner as the routine field samples?				
Part 5: Sample Management				
5.1 Collection				
Are the samples taken from a representative point of the source?				
Are the samples homogenous where appropriate?				
When possible, does sampling originate from the suspected least contaminated location first and progress to the suspected most contaminated location?				
Are samples for different analyte groups collected in the appropriate order?				
Are samples collected for all required analyses?				
Are samples to be tested for dissolved metals filtered prior to preservation?				
Is every effort made to prevent cross-contamination of samples?				
Are gloves worn by all samplers handling purging equipment, sampling equipment, measurement equipment, and sample containers?				
Are new, clean unpowdered gloves used for each glove change?				
Is care taken to avoid contact with sample and sample container interiors?				
Are VOC sample containers protected from any fuel sources and fuel-powered equipment?				
Do VOC sample containers remain capped until just prior to sample collection and do they remain capped after sample collection?				
Where applicable, are samples collected for measurement of dissolved components, filtered, preserved with acid, and placed on ice within 15 minutes of collection?				

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5.2 Collection Devices				
Is sample collected using an intermediate collection device?				
Are intermediate collection devices rinsed with ample amounts of site water prior to collecting the sample?				
Is rinse water from intermediate devices discarded away from and downstream of the sampling location?				
Is the use of intermediate collection devices avoided when sampling for VOC's, oil and grease, or microbiologicals, where practical?				
Are any intermediate collection devices constructed of material appropriate for the analytes to be measured?				
Are sample containers submerged neck first, inverted into the oncoming direction of flow where applicable, slowly filled, and returned to the surface for preservation, if applicable?				
5.3 Sample Labeling				
Is each sample container or group of containers tagged or labeled with a unique field identification code that distinguishes the sample from all other samples?				
Are the unique identification codes for samples recorded in a manner that links the codes to all other field records associated with the samples?				
Is waterproof indelible ink used to label containers?				
5.4 Storage				
Are samples for different parameters segregated during storage?				
Are samples stored on ice?				
Is the cooler clearly labeled?				
Are samples properly preserved (if applicable)?				
5.5 Preservation				
Do all sample preservation techniques conform to SOP or method requirements?				
Are all samples properly preserved within 15 minutes, as applicable?				
Are the preparation and dispensing of preservatives documented and traceable?				
Is preservation information and verification recorded for each sample, as applicable?				
Are samples placed on ice immediately after collection, if applicable?				
If the samples are found to contain cyanide, are they NOT acidified?				
5.6 Delivery				
Are samples protected during delivery to prevent breakage?				
Are samples shipped in a timely manner?				

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5.7 Disposal				
Are wastes generated as a result of the sampling project containerized and stored for proper disposal according to applicable local, state, and federal regulations?				
Are all sampling-derived waste containers properly labeled?				
Is all sampling-derived waste properly disposed of?				
5.8 Documentation				
Is waterproof ink used for all paper documentation?				
Are the date and time of sample collection recorded for all samples?				
Are the ambient field conditions recorded for all samples?				
Is a specific description of each sampling location (source) recorded?				
Does the chain of custody/traffic report include the following: date, time, sample numbers, sampler names, shipping method, number of samples, matrix, and comments?				
Is preservation information recorded on the chain of custody/traffic report?				
Are copies of traffic reports or COC sent to the proper recipients?				
Are deviations, additions, or exclusions from the documented sampling procedure recorded in detail with the associated sampling information?				
Are these deviations included in all documents containing environmental test and/or calibration results?				
Are these deviations communicated to the appropriate personnel?				
Are all errors in documentation (if applicable) corrected and initiated without obliteration?				
5.9 Field Reagents				
Are the concentration (or other assay value), the vendor catalog number and the description of the standard or reagent recorded for all preformulated solutions, neat liquids, powders, and blank water?				
Are certificates of assay, grade and other vendor specifications for all standards and reagents retained and recorded for the standards and reagents?				
Are the lot numbers and inclusive dates of use recorded for all reagents, detergents, solvents, and other chemicals used for decontamination and preservation of samples?				
Are the expiration dates for all calibration standards and reagents recorded?				
Are expired standards and reagents verified prior to use during sample collection?				
Are all steps used for preparation of standards or reagents in-house documented either by description or reference to an SOP?				

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Part 6: Field Analyses				
6.1 General Field Test Information				
Are all field measurement tests and related data recorded and linked to the project, the date, and the sample source?				
Are all field measurements recorded with the appropriate units, the value of the test result, the parameter measured, the name of the analyst performing the test, the time of the measurement and the unique identification for the test instrument used?				
6.2 pH				
Are all samples requiring pH adjustment tested for proper pH preservation?				
Is at least one sample per analyte group requiring pH adjustment tested for proper preservation during repeat sampling?				
Is pH paper or a pH electrode inserted into sample containers?				
Do the pH meter and electrode system meet SOP specifications for accuracy, reproducibility and design?				
Are all measurements corrected for temperature (manual or automatic)?				
Is a pH 7 buffer used as the first calibration standard?				
For pH, do all calibration verifications meet the acceptance criteria?				
If the calibration and/or calibration verifications did not meet the acceptance criteria, is the calibration or verification identified as a failure and was this documented in the calibration log?				
Are all sample measurements associated with acceptable calibration verifications?				
Is the pH meter system checked on a weekly basis to ensure >90% theoretical electrode slope?				
Are the field instrument probes rinsed with deionized or distilled water between standard solutions and between sample measurements?				
Are instrument pH readings allowed to stabilize before pH values are recorded?				
6.3 Filtration				
Are samples collected for analysis of dissolved components filtered within 15 minutes of collection and before addition of chemical preservatives where appropriate?				
Unless otherwise specified, are applicable samples filtered using a 0.45-μm pore size?				
6.4 Temperature				
Do the temperature measurement devices meet SOP and/or sampling event specifications for design and measurement resolution?				

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Are all sample measurements associated with calibration verifications of the temperature measurement device at a minimum of two temperatures using a NIST-traceable thermometer?				
If the calibration and/or calibration verifications did not meet the acceptance criteria, is the calibration or verification identified as a failure and was this documented in the calibration log?				
Are all temperature measurements chronologically associated with acceptable calibration verifications?				
Are the temperature device readings allowed to stabilize before measurement values were recorded?				
6.5 Conductivity				
Do the specific conductance meter and electrode system meet the SOP and/or sampling event specifications for accuracy and reproducibility?				
Do all calibration verifications meet the acceptance criterion?				
If the calibration and/or calibration verifications did not meet the acceptance criteria, is the calibration or verification identified as a failure and was this documented in the calibration log?				
Are all conductivity measurements chronologically associated with acceptable calibration verifications?				
Are all conductivity measurements corrected for temperature (manual or automatic)?				
Is the instrument allowed to stabilize before measurement values are recorded?				
6.6 Turbidity				
Does the turbidimeter meet the SOP and/or sampling event specifications for accuracy and reproducibility?				
Are all sample measurements associated with acceptable calibration verifications?				
If the calibration and/or calibration verifications did not meet the acceptance criteria, is the calibration or verification identified as a failure and was this documented in the calibration log?				
Are the sample cells (optical cuvettes) inspected for scratches and discarded or coated with a silicone oil mask, as necessary?				
Are the sample cells (optical cuvettes) optically matched for calibrations and sample measurements?				
Are the sample cells (optical cuvettes) cleaned with detergent and deionized or distilled water between standard solutions and between sample measurements, as applicable?				
Are the sample cells (optical cuvettes) rinsed with sample prior to filling with sample for measurement?				

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Is the exterior of the sample cell (optical cuvette) kept free of fingerprints and dried with a lint-free wipe prior to insertion in the turbidimeter?				
6.7 Dissolved Oxygen				
Do the dissolved oxygen meter and electrode system meet the SOP and/or sampling event specifications for accuracy and reproducibility?				
Are all sample measurements associated with acceptable calibration verifications?				
If the calibration and/or calibration verifications did not meet the acceptance criteria, is the calibration or verification identified as a failure and was this documented in the calibration log?				
Are all measurements corrected for temperature (manual or automatic)?				
Are all measurements corrected for salinity, where applicable (manual or automatic)?				
Is the salinity (conductivity) sensor calibration verified?				
Is the dissolved oxygen electrode stored in a water-saturated air environment when not in use?				
Are the dissolved oxygen readings allowed to stabilize before measurement values were recorded?				